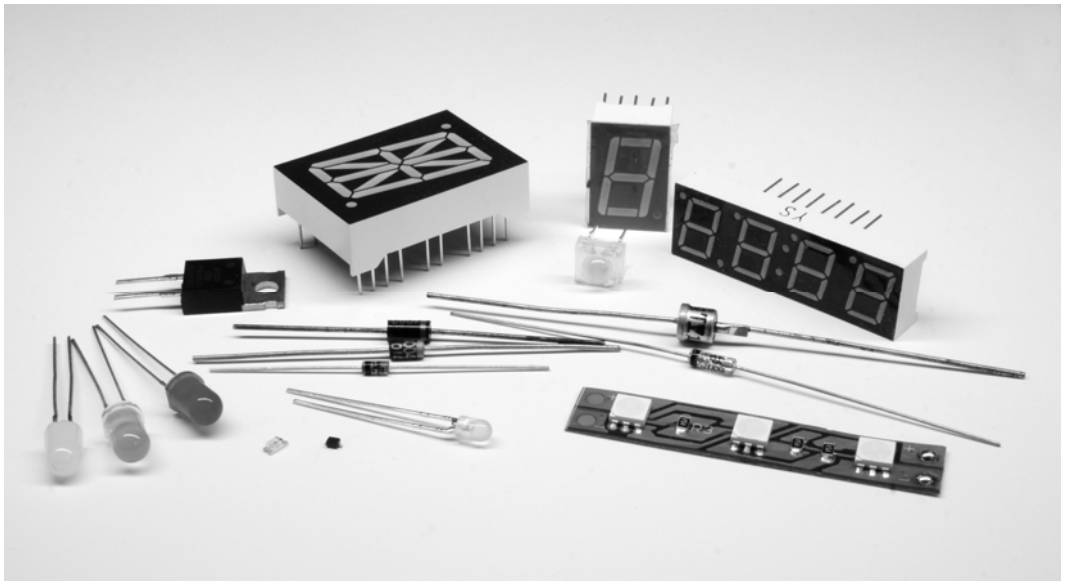


Figure 12-5. *Diodes*

There are generally two types of diodes that we will most often use. The first is a general purpose rectifier diode, such as the 1N4001. This is used for applications where we need to protect our circuit from reverse voltages that can be caused by inductive loads such as motors, solenoids, or relays.

Arguably the most common diode is the light-emitting diode that we have used many times throughout the book already. LEDs are fairly popular with the microcontroller crowd because they are so simple to connect and require small amounts of power to run. These diodes will emit light when electricity passes through them. The anode or positive (+) pin is usually the longer pin, while the cathode or negative (-) pin is the shorter one. LEDs come in many shapes, sizes, and colors, from 10mm to 3mm; miniature surface mount to large multi-watt LEDs; as well as seven-segment and alphanumeric displays for displaying numbers and letters.

Transistors

Transistors can be used just like an electrical switch. A small amount of current supplied to one pin of the transistor can switch a much greater load connected to the other pins. A bipolar transistor has three pins or wires that include the base, collector, and emitter. Figure 12-6 shows a few of the different packages available for transistors.